DOCKHUB · PRODUCT REQUIREMENTS AND SPECIFICATIONS

DockHub product requirements and specifications

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.9	2026-10-31	Amish Chadha	Migrated at the kit 1.9.1 upgrade: R1 to R16 became PR-001 to PR-016 in the PRS columns with a Was column, every PR traced to a user requirement in DKH-URS-001; PR-017 added for the aerosol fan rule (DKH-DEC-001 D1); targets, wording and status unchanged
0.8	2026-10-02	Amish Chadha	Status from DKH-CAL-001 v0.6 after the approved follow-ups: R15 met with the anchors and site pad chosen; R16 USD 1,519 with them; R12 token list with anonymous prepaid cards
0.7	2026-10-02	Amish Chadha	R6 restated as about 9 % and R9 as full-rate charging up to about 30 °C ambient (DKH-DEC-001); status updated
0.6	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.5	2026-10-01	Amish Chadha	

			Status from DKH-CAL-001 v0.3 after the constructable design (DKH-DDR-003); R16 not met
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3. R6 relaxed to 10 % and R16 redefined to the two-bay fit (DKH-DDR-001); R2 to SwapCell interface v0.3; status from DKH-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-25	Amish Chadha	Initial scaffold

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What the station must do, each requirement traced to a user need in DKH-URS-001 (links in DKH-RTM-001). The requirements describe option A, swap bays in a steel cabinet, the current design, which is carried forward unchanged; the concept options are scored in DKH-OPT-001 and the choice is proposed, awaiting Amish. They were numbered R1 to R16 before kit 1.8; the **Was** column keeps the old numbers, which the calculation note (DKH-CAL-001) and the build plan still use. They are checked by calculation in DKH-CAL-001 (TRL 3) and are not yet validated with riders, hosts or a city (DKH-PRB-001). On 2026-09-25 Amish accepted the review recommendations (DKH-DDR-001, DKH-DDR-002): PR-006 relaxed from 20 % to 10 %, PR-016 covers the four-bay cabinet with two bays fitted, and PR-002 cites SwapCell interface v0.3. On 2026-10-02 Amish restated PR-006 as about 9 % and PR-009 as full-rate charging up to about 30 C ambient (DKH-DEC-001).

On paper, 12 of 17 are met or met by design, with qualifications from the independent review (PR-005, PR-006, PR-009, PR-015, PR-017). PR-016 (cost) is over its value-engineering target; PR-008 (fire containment) and PR-009 (climate) are at risk; PR-007 (fire detection) and PR-014 (maintainability) cannot be verified until hardware exists. The STEM check (DKH-STM-001) adds PR-017.

The **design duty case** used throughout is 20 swaps a day, each returning a SwapCell pack at about 15 % state of charge, in a street location with 4.5 peak sun hours and ambient air between -10 and 45 C. Because a swap needs an empty bay, a station with *n* bays holds *n* - 1 packs.

Table 1. Product requirements and specifications. Status at TRL 3 from DKH-CAL-001 (python docs/04-calcs/sizing.py).

ID	Was	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
PR-001	R1	UR-001	Performance	Fast swap	60 s or less from tapping the reader to closing the second door, for a rider familiar with the station	60	max	s	Task analysis; later timed trials with riders	Met (31 s)
PR-002	R2	UR-005	Interface	SwapCell compatibility	Every bay builds to SwapCell interface v0.3: envelope, guide faces, blind-mate receptacle with the 10 kΩ INTERLOCK coding resistor, class D catch, CAN heartbeat and message set; no change to the interface	v0.3	exact	n/a	Interface review with the SwapCell project	Met by design review; one CAN channel per bay
PR-003	R3	UR-002	Performance	Charge speed	20 to 80 % in 1.5 h or less; full charge in 2.5 h or less, per bay, at 25 °C	1.5 (20 to 80 %); 2.5 (full)	max	h	Charge profile calculation	Met (1.2 h and 2.3 h)
PR-004	R4	UR-002	Performance	Throughput	20 or more swaps a day from 4 bays at the design duty case	20	min	swaps a day	Day model with a demand profile	Met by the four-bay fit (20 of 20 served, longest wait 4 min; ceiling 36 a day). The two-bay first prototype serves 9 of 20
PR-005	R5	UR-009	Logistics	Grid supply	One single-phase branch circuit (120 V	1.5	max	kW	Load calculation	Met (1.25 kW; 11.0 A at 120 V) with

ID	Was	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
					15 A or 230 V 10 A); peak draw 1.5 kW or less					chargers of power factor 0.9 or better Independent review (DKH-STM-001): at 10 % low mains voltage constant-power chargers draw 12.2 A at 120 V, over the 12 A continuous limit if all four run; the simulated day runs at most 3 (8.3 A); low-voltage case to add to DKH-CAL-001
PR-006	R6	UR-013	Performance	Solar contribution	Canopy solar supplies about 9 % of daily charging energy at the design duty case, with normal routing (restated on 2026-10-02, DKH-DEC-001; relaxed from 20 % to 10 % by DKH-DDR-001)	9	about	% of charging energy	Day model	Met on paper, thin margin (9.3 %; 11.4 % if flat packs are routed to bay 1 in daylight, which turns one rider in 20 away, so normal routing is kept; 16.1 % if the solar output is later switched to the flattest pack, a TRL 4 trial) Basis: 9.3 % is measured at the pack terminals; at the inputs (0.77 kWh panel against 8.04 kWh grid to chargers) it

ID	Was	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
										is 8.7 %, on one clear design day; the definition is proposed, awaiting Amish (DKH-DEC-001 O6)
PR-007	R7	UR-003	Safety	Fire detection and response	Detect a pack or bay over 60 °C, or smoke, within 10 s; open that bay's charge contactor within 1 s of detection; sound a local alarm and send a remote alert; vent gas to the rear roof louver, away from the user side	10 detect; 1 contactor	max	s	Latency budget; later bench test with a heat source	Not verifiable at TRL 3 (electronic latency 3.1 s, contactor 0.15 s)
PR-008	R8	UR-003	Safety	Fire containment	A thermal runaway in one bay does not spread to a neighboring bay or pack for 30 min or more, and flame does not leave the cabinet front	30	min	min	Later propagation test by an accredited lab	**At risk**; not verifiable at TRL 3
PR-009	R9	UR-004	Environment and durability	Climate	Enclosure IP54 or better; operate (store and swap) at -10 to 45 °C ambient; full-rate charging up to about	IP54; -10 to 45; 30	min IP; range; about	code; C; C	Thermal calculation; later chamber test	**At risk**: uninterrupted charging needs ambient below about 29 to 39 °C (pack charge limit 45 °C), so the

ID	Was	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
					30 °C ambient, with daytime pauses accepted above it and no charging start for cold-soaked packs below about -1 °C; never charge a pack outside the pack's reported charge temperature window (restated on 2026-10-02, DKH-DEC-001)					warmest bay may pause just under 30 °C; cold-soaked packs cannot start below about -1 °C, as now stated IP54 is not demonstrated: the vent slots and filtered intake make it doubtful until an enclosure test (STEM S27); the lowest calculated limit for uninterrupted charging is 29.4 C (S9)
PR-010	R10	UR-007	Security	Security	1.5 mm or thicker steel; fail-secure bay locks; cabinet anchored to its pad with tamper-resistant fixings; a pack is released only against a valid token and a returned pack	1.5	min	mm	Design review	Met by design review
PR-011	R11	UR-008	Usability	Accessibility and use	All operable parts (reader, buttons, bay handles) 0.38 to 1.22 m (15 to 48 in) above the ground;	0.38 to 1.22	range	m	Model check; later user trials	Met (0.73 to 1.10 m)

ID	Was	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
					status shown by light and symbol as well as text; usable with gloves and at night					
PR-012	R12	UR-010	Privacy	Privacy and data	No cameras or microphones; logs hold pack ID, bay, time, energy and state of health, with the rider as an opaque token; operates offline for 24 h or more and syncs later	0 cameras; 24	min	h offline	Design review	Met by design review (token list of 200 kB, including anonymous prepaid cards with a pack deposit)
PR-013	R13	UR-009	Logistics	Sidewalk footprint	Cabinet footprint 1.0 x 0.6 m or less; canopy underside 2.1 m or more above the sidewalk	1.0 x 0.6; 2.1	max; min	m; m	Model check	Met (1.0 x 0.5 m with a flush plinth; lowest canopy point over the sidewalk 2.17 m)
PR-014	R14	UR-011	Maintenance	Maintainability	Any charger, lock, bay cradle or controller replaced in 15 min or less by one technician with hand tools, through the service door	15	max	min	Design review; later timed trial	Not verifiable at TRL 3
PR-015	R15	UR-006	Safety	Wind and anchoring	Stays upright and intact at a 30 m/s gust	30	min	m/s gust	Wind load calculation	Met on paper: four M12 stainless wedge

ID	Was	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
					with the canopy fitted					anchors (3.8 kN design tension each) into a 1,400 x 1,100 x 300 mm cast pad of about 1,109 kg; equilibrium factor 1.12 at a 30 m/s gust. The anchor rating is confirmed from the maker's data when bought Upright only; STEM S16 is plausible (6 % margin; anchor design tension 4.13 kN with matching factors, proposed 4.2 kN, DKH-DEC-001 O5). The canopy being intact (panel clamps, roof pull-through at 1,251 N per post) and wind from other directions are not yet checked
PR-016	R16	UR-012	Cost	Cost	Parts 1,200 USD or less for the four-bay cabinet with two bays fitted (first prototype), SwapCell	1,200	max	USD	Priced BOM (`bom/bom.csv`)	**Over the value-engineering target by $319** ($1,519 against $1,200: $1,214 for the buildable

ID	Was	Traces to	Category	Title	Requirement	Target	Tolerance	Units	Verification	Status
					packs excluded (redefined, DKH-DDR-001)					cabinet, DKH-DDR-003, plus $305 for the anchors and site pad chosen on 2026-10-02; $1,743 with all four bays fitted)
PR-017	new	UR-003	Safety	Aerosol is kept in	On a condensed-aerosol discharge in a bay the controller shall stop both roof fans and leave the plenum to the passive hood slot (240 cm2 free area), and restart the fans only on a technician's reset; PR-007's vent path to the roof is then the passive hood slot (DKH-DEC-001 D1, from DKH-STM-001 S13)	1 fan-off rule	Pass or fail	n/a	Design review; retention test in a bay mock-up (TRL 4)	Met by design (control rule); retention time to measure at TRL 4 (S13); gas build-up with the fans stopped is not yet assessed (S29, DKH-DEC-001 O4, proposed, awaiting Amish)

Assumptions

- SwapCell pack (interface v0.3): 46.8 V nominal, about 468 Wh, 2.85 kg, charge on a certified 54.6 V 5 A charger in 2.3 h to full and 1.2 h for 20 to 80 %, charge window 0 to 45 °C (SWC-PRC-001 v0.3, SWC-CAL-001).
- Charger efficiency about 90 %, cell charge efficiency about 95 %, cabinet auxiliary load about 30 W on average and 40 W peak.
- 400 W panel, 4.5 peak sun hours, 0.8 derate, 96 % MPPT, feeding bay 1 only.

- Packs are released at 80 % state of charge or more, fullest first; riders arrive with a lunch and a dinner peak and leave after waiting 15 min (DKH-CAL-001 Table 1).
- Delivery riders cover roughly 60 to 100 km a day at about 10 Wh/km. This is an assumption to check with riders; it implies 1 to 2 swaps per rider per shift.
- Accessibility reach range from the US ADA forward reach limits (15 to 48 in); local rules may differ.